

Problem Set Lab 01 (graded), Sep. 03, 2026 (Mathematical Foundation of Machine Learning)

Goals. The goals of this lab are to:

- Familiarize yourself with the mathematical foundations of the machine learning course.

Submission instructions:

- Please submit a PDF file to canvas.
- Deadline: 23.59 on Sep. 10, 2026

Review of Linear Algebra

Problem 1 (Orthogonal Projections and Least Squares (10 pts)):

Let $V \subseteq \mathbb{R}^n$ be a subspace, and let $P \in \mathbb{R}^{n \times n}$ be the matrix of the orthogonal projection onto V , i.e., $\text{range}(P) = V$, $P^2 = P$, and $P^\top = P$. For $x \in \mathbb{R}^n$ and any $y \in V$, prove that

$$\|x - Px\|_2 \leq \|x - y\|_2, \quad (1)$$

with equality if and only if $y = Px$. (This is the geometric heart of least-squares regression: among all predictions lying in the span of the features, the projection is the closest one.)

Problem 2 (Gradient of the Squared Loss and the Normal Equations (10 pts)):

Let $X \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$, and consider the squared loss

$$f(w) = \|Xw - y\|_2^2, \quad w \in \mathbb{R}^d. \quad (2)$$

1. Show that $\nabla f(w) = 2X^\top(Xw - y)$.
2. Show that f attains its minimum exactly at the solutions w^* of the normal equations $X^\top Xw^* = X^\top y$, and that such a w^* always exists.
3. Show that the minimizer is unique if and only if $\text{rank}(X) = d$ (i.e., the columns of X are linearly independent).

Problem 3 (Positive Semidefinite Matrices and Square Roots (10 pts)):

Let $A \in \mathbb{R}^{n \times n}$ be symmetric. We say A is positive semidefinite (PSD), written $A \succeq 0$, if $x^\top Ax \geq 0$ for all $x \in \mathbb{R}^n$.

1. Prove that $A \succeq 0$ if and only if all eigenvalues of A are nonnegative.
2. Prove that if $A \succeq 0$, there exists a symmetric PSD matrix B with $B^2 = A$. Moreover, show B commutes with every matrix that commutes with A .
3. Show that for any matrix $M \in \mathbb{R}^{m \times n}$, the Gram matrix $M^\top M$ is PSD, and $\text{rank}(M^\top M) = \text{rank}(M)$.

(PSD matrices appear throughout ML as covariance matrices, kernel (Gram) matrices, and Hessian matrices of convex losses.)

Problem 4 (Sylvester's Determinant Identity (10 pts)):

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$.

1. Prove that

$$\det(I_m + AB) = \det(I_n + BA). \quad (3)$$

(Hint: compute the determinant of the block matrix $\begin{pmatrix} I_m & A \\ B & I_n \end{pmatrix}$ by two different sequences of row and column operations.)

2. Deduce that for every $\lambda \in \mathbb{R}$ (assume $m \geq n$ for simplicity),

$$\det(\lambda I_m - AB) = \lambda^{m-n} \det(\lambda I_n - BA), \quad (4)$$

and conclude that AB and BA have exactly the same nonzero eigenvalues, counted with algebraic multiplicity.

3. Prove that for every integer $k \geq 1$,

$$\text{tr}((AB)^k) = \text{tr}((BA)^k). \quad (5)$$

Problem 5 (Sylvester and Frobenius Rank Inequalities (10 pts)):

1. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. Prove *Sylvester's rank inequality*:

$$\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n. \quad (6)$$

2. Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, $C \in \mathbb{R}^{p \times q}$. Prove the *Frobenius rank inequality*:

$$\text{rank}(AB) + \text{rank}(BC) \leq \text{rank}(B) + \text{rank}(ABC). \quad (7)$$

3. Deduce from (a) that if $AB = 0$, then $\text{rank}(A) + \text{rank}(B) \leq n$, and give an example of a pair (A, B) with $AB = 0$ for which equality holds.

Review of Probability Theory

Problem 6 (Cauchy–Schwarz Inequality for Random Variables (10 pts)):

Let X and Y be random variables with $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[Y^2] < \infty$. Prove that

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2] \mathbb{E}[Y^2], \quad (8)$$

and characterize when equality holds. (Taking $Y = \mathbf{1}$ recovers $\text{Var}(X) \geq 0$; the equality case is the basis of the linear least-squares predictor of Y from X .)

Problem 7 (Jensen's Inequality and Maximum Likelihood (10 pts)):

1. Let X be a positive random variable with $\mathbb{E}[X] < \infty$. Prove that

$$\mathbb{E}[\log X] \leq \log \mathbb{E}[X]. \quad (9)$$

2. Use part (a) to show that for any two probability mass functions p and q on a finite set $\mathcal{X}$ with $q(x) > 0$ whenever $p(x) > 0$, the Kullback–Leibler divergence satisfies

$$D_{\text{KL}}(p \parallel q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)} \geq 0, \quad (10)$$

with equality if and only if $p = q$.

(Part (b) is why maximum likelihood works: the expected log-likelihood is maximized by the true distribution.)

Problem 8 (Markov's Inequality and the Union Bound (10 pts)):

1. Let X be a nonnegative random variable and $t > 0$. Prove Markov's inequality:

$$P(X \geq t) \leq \frac{\mathbb{E}[X]}{t}. \quad (11)$$

2. Let $X_1, \dots, X_n$ be i.i.d. nonnegative random variables with finite mean μ . Using part (a) and the union bound, prove that

$$P\left(\max_{1 \leq i \leq n} X_i \geq t\right) \leq \frac{n\mu}{t}. \quad (12)$$

3. Suppose additionally that $\mathbb{E}[X^2] < \infty$. Show that $\frac{1}{n} \max_{1 \leq i \leq n} X_i \rightarrow 0$ in probability as $n \rightarrow \infty$, i.e., for every $\varepsilon > 0$,

$$P\left(\frac{1}{n} \max_{1 \leq i \leq n} X_i \geq \varepsilon\right) \rightarrow 0. \quad (13)$$

(“A sample average cannot be dominated by a single lucky outlier”—the union bound is the workhorse behind empirical-process arguments in learning theory.)

Problem 9 (The Law of Rare Events (10 pts)):

1. Let $\lambda > 0$ be fixed, and let $X_n \sim \text{Binomial}(n, \lambda/n)$. Prove that for every fixed integer $k \geq 0$,

$$\lim_{n \rightarrow \infty} P(X_n = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad (14)$$

i.e., X_n converges in distribution to a Poisson random variable with parameter λ .

2. Let $N \sim \text{Poisson}(\lambda)$. Prove that

$$P(N \text{ is even}) = \frac{1 + e^{-2\lambda}}{2}. \quad (15)$$

Problem 10 (The Minimum of Independent Exponential Random Variables (10 pts)):

Let $X_1, \dots, X_n$ be independent random variables with $X_i \sim \text{Exp}(\lambda_i)$ (rate $\lambda_i > 0$, i.e., $P(X_i > t) = e^{-\lambda_i t}$ for $t \geq 0$). Define

$$M = \min_{1 \leq i \leq n} X_i, \quad I = \arg \min_{1 \leq i \leq n} X_i \quad (16)$$

(the index of the smallest X_i ; it is uniquely determined with probability 1).

1. Show that $M \sim \text{Exp}(\Lambda)$, where $\Lambda = \lambda_1 + \dots + \lambda_n$.
2. Show that $P(I = k) = \frac{\lambda_k}{\Lambda}$ for each k .
3. Show that I and M are independent.